

## 7.5 Commutators and Trace-Zero Matrices

The expression  $\mathbf{AB} - \mathbf{BA}$  has come up several times already. In Chapter 0 its trace was 0, which showed that  $\mathbf{AB} - \mathbf{BA} = \mathbf{I}_n$  is impossible over  $\mathbb{R}$  (Exercise (C1:  $\mathbf{AB} - \mathbf{BA} = \mathbf{I}$  is impossible, or is it?)). In Chapter 3 the same trace argument showed that the relation  $DS - SD = \text{id}$  between differentiation and multiplication by  $x$  on  $F[x]$  has no finite-dimensional analogue in characteristic 0 (Exercise (C1: The Heisenberg relation)). Both arguments use one direction only: every matrix of the form  $\mathbf{AB} - \mathbf{BA}$  has trace zero. This section asks whether the converse holds. Is every trace-zero matrix of the form  $\mathbf{AB} - \mathbf{BA}$ ? The answer is yes over a field of characteristic 0, and the proof is a good example of the block techniques of this chapter: move the matrix into a convenient shape by a similarity, one row and column at a time, and then solve an equation that has become entrywise.

**Which fields.** Commutators, their basic rules, the first similarity lemma and the final theorem on spans work over **every** field  $F$ . The lemma on zero diagonals and Shoda's theorem assume that  $F$  has **characteristic** 0. We point out exactly where that hypothesis is used, and what happens in characteristic  $p$ .

### 7.5.1 Commutators

Two matrices  $\mathbf{A}$  and  $\mathbf{B}$  commute when  $\mathbf{AB} = \mathbf{BA}$ , and most pairs do not. The difference  $\mathbf{AB} - \mathbf{BA}$  measures the failure, and it keeps recurring: in the trace arguments above, in the relation between differentiation and multiplication by  $x$ , and in the description of trace-like functionals (Exercise (C2: The trace is the only trace-like functional)). So we give it a name.

*The commutator of two matrices is the amount by which they fail to commute.*

#### Definition 7.5.1 — Commutator

Let  $n \geq 1$  and  $\mathbf{A}, \mathbf{B} \in M_n(F)$ . The **commutator** of  $\mathbf{A}$  and  $\mathbf{B}$  is

$$[\mathbf{A}, \mathbf{B}] := \mathbf{AB} - \mathbf{BA} \in M_n(F).$$

A matrix  $\mathbf{C} \in M_n(F)$  is a **commutator** if  $\mathbf{C} = [\mathbf{X}, \mathbf{Y}]$  for **some**  $\mathbf{X}, \mathbf{Y} \in M_n(F)$ . For operators  $S, T \in \mathcal{L}(V)$  on a vector space  $V$ , we write likewise  $[S, T] := ST - TS \in \mathcal{L}(V)$ .

In words:  $[\mathbf{A}, \mathbf{B}]$  is “ $\mathbf{A}$  then  $\mathbf{B}$ ” minus “ $\mathbf{B}$  then  $\mathbf{A}$ ”, in that order, and it is the zero matrix exactly when  $\mathbf{A}$  and  $\mathbf{B}$  commute. The second sentence of the definition is about a single matrix: being a commutator means that **there exist** two matrices whose commutator it is, and it says nothing about which ones.

**Examples.**

1. **Degenerate cases.**  $[\mathbf{A}, \mathbf{A}] = 0$  and  $[\mathbf{A}, c\mathbf{I}_n] = c\mathbf{A} - c\mathbf{A} = 0$  for every  $c \in F$ . So the zero matrix is a commutator, in many ways. In  $M_1(F) = F$  every commutator is  $ab - ba = 0$ , so for  $n = 1$  the only commutator is 0; this is the smallest case, and it already matches the trace condition, since the only  $1 \times 1$  matrix of trace 0 is (0).
2. **Matrix units.** Recall that  $\mathbf{E}_{ij}\mathbf{E}_{kl}$  is  $\mathbf{E}_{il}$  if  $j = k$  and 0 otherwise (Definition 0.9.9). So

for  $i \neq k$ ,

$$[\mathbf{E}_{ij}, \mathbf{E}_{jk}] = \mathbf{E}_{ik} - 0 = \mathbf{E}_{ik}, \quad \text{and} \quad [\mathbf{E}_{ij}, \mathbf{E}_{ji}] = \mathbf{E}_{ii} - \mathbf{E}_{jj} \quad (i \neq j).$$

For instance  $[\mathbf{E}_{12}, \mathbf{E}_{21}] = \text{diag}(1, -1)$  in  $M_2(F)$ , and  $[\mathbf{E}_{11}, \mathbf{E}_{12}] = \mathbf{E}_{12}$ . Both have trace 0.

3. **Operators on an infinite-dimensional space.** On  $F[x]$ , with  $D$  differentiation and  $S$  multiplication by  $x$ , Exercise (C1: The Heisenberg relation) (a) shows  $[D, S] = \text{id}$ . So the identity is a commutator of operators on  $F[x]$ . This does not contradict anything, because there is no trace on  $\mathcal{L}(F[x])$ : the trace was defined only on finite-dimensional spaces (Definition 3.10.2).

**Non-example by minimal change.** Change the minus sign to a plus:  $\mathbf{AB} + \mathbf{BA}$ . It is still linear in  $\mathbf{A}$  and in  $\mathbf{B}$ , and it is still built from the two products  $\mathbf{AB}$  and  $\mathbf{BA}$ . What is lost is the trace:  $\text{tr}(\mathbf{AB} + \mathbf{BA}) = 2 \text{tr}(\mathbf{AB})$ , which has no reason to vanish. For example  $\mathbf{E}_{12}\mathbf{E}_{21} + \mathbf{E}_{21}\mathbf{E}_{12} = \mathbf{E}_{11} + \mathbf{E}_{22} = \mathbf{I}_2$ , with trace 2. It also no longer detects commuting:  $\mathbf{AA} + \mathbf{AA} = 2\mathbf{A}^2$  is non-zero for  $\mathbf{A} = \mathbf{I}_n$  over  $\mathbb{Q}$ , although  $\mathbf{A}$  commutes with itself.

**Why this definition.** The order in  $\mathbf{AB} - \mathbf{BA}$  is a convention; the other order gives  $[\mathbf{B}, \mathbf{A}] = -[\mathbf{A}, \mathbf{B}]$ , which is a commutator too. The notion is invariant under a change of basis: a similarity applied to  $\mathbf{X}$  and  $\mathbf{Y}$  applies to  $[\mathbf{X}, \mathbf{Y}]$ . So “being a commutator” is a property of the operator  $\mathbf{x} \mapsto \mathbf{C}\mathbf{x}$ , not of the basis used to write  $\mathbf{C}$ , and we are free to choose a convenient basis. This is what the proof of the main theorem exploits.

The basic rules are collected here.

### Proposition 7.5.1 — Rules for Commutators

Let  $\mathbf{A}, \mathbf{A}', \mathbf{B} \in M_n(F)$ ,  $c \in F$ , and let  $\mathbf{P} \in M_n(F)$  be invertible. Then:

- (a)  $[\mathbf{A} + \mathbf{A}', \mathbf{B}] = [\mathbf{A}, \mathbf{B}] + [\mathbf{A}', \mathbf{B}]$ ,  $[c\mathbf{A}, \mathbf{B}] = c[\mathbf{A}, \mathbf{B}]$ , and  $[\mathbf{B}, \mathbf{A}] = -[\mathbf{A}, \mathbf{B}]$ ; in particular the commutator is linear in each slot.
- (b)  $\mathbf{P}^{-1}[\mathbf{A}, \mathbf{B}]\mathbf{P} = [\mathbf{P}^{-1}\mathbf{A}\mathbf{P}, \mathbf{P}^{-1}\mathbf{B}\mathbf{P}]$ .
- (c)  $\text{tr}[\mathbf{A}, \mathbf{B}] = 0$ .

*Proof.*

- (a) By Theorem 0.9.4,  $(\mathbf{A} + \mathbf{A}')\mathbf{B} - \mathbf{B}(\mathbf{A} + \mathbf{A}') = (\mathbf{AB} - \mathbf{BA}) + (\mathbf{A}'\mathbf{B} - \mathbf{BA}')$  and  $(c\mathbf{A})\mathbf{B} - \mathbf{B}(c\mathbf{A}) = c(\mathbf{AB} - \mathbf{BA})$ . The last identity is  $\mathbf{BA} - \mathbf{AB} = -(\mathbf{AB} - \mathbf{BA})$ . Linearity in the second slot follows from the first slot and antisymmetry.
- (b) Inserting  $\mathbf{PP}^{-1} = \mathbf{I}_n$ ,  $\mathbf{P}^{-1}\mathbf{A}\mathbf{B}\mathbf{P} = (\mathbf{P}^{-1}\mathbf{A}\mathbf{P})(\mathbf{P}^{-1}\mathbf{B}\mathbf{P})$ , and likewise for  $\mathbf{BA}$ . Subtract, using distributivity.
- (c) By Theorem 0.9.6 (1) and (3),  $\text{tr}(\mathbf{AB} - \mathbf{BA}) = \text{tr}(\mathbf{AB}) - \text{tr}(\mathbf{BA}) = 0$ .

■

So every commutator lies in the subspace of trace-zero matrices of Example (Trace-Free Matrices). The question of this section is whether it fills that subspace.

## 7.5.2 The identity in characteristic $p$

Rule (c) is the whole content of the Heisenberg exercises: if  $[\mathbf{X}, \mathbf{Y}] = \mathbf{I}_n$ , then  $n \cdot 1 = \text{tr } \mathbf{I}_n = 0$ , which is impossible when  $F$  has characteristic 0 (Definition 0.7.2). The obstruction is the number  $n \cdot 1$ , not the identity matrix itself. In characteristic  $p$ , the  $p \times p$  identity has trace  $p \cdot 1 = 0$ , and the obstruction disappears. In fact the pair  $D, S$  of Example 3 survives in a finite form.

### Example. (The Identity as a Commutator in Characteristic $p$ )

Let  $F$  have prime characteristic  $p$  (for instance  $F = \mathbb{F}_p$ ). Let  $\mathbf{D}_p \in M_p(F)$  have  $(k, k+1)$ -entry  $k \cdot 1$  for  $1 \leq k \leq p-1$ , and let  $\mathbf{S}_p \in M_p(F)$  have  $(k+1, k)$ -entry 1 for  $1 \leq k \leq p-1$ , all other entries 0. Show that  $[\mathbf{D}_p, \mathbf{S}_p] = \mathbf{I}_p$ .

*Solution.* The columns of these matrices describe differentiation and multiplication by  $x$  on the list  $1, x, \dots, x^{p-1}$ , where  $x^p$  is replaced by 0. By Theorem 0.9.2,  $\mathbf{S}_p \mathbf{e}_k = \mathbf{e}_{k+1}$  for  $k < p$  and  $\mathbf{S}_p \mathbf{e}_p = \mathbf{0}$ , while  $\mathbf{D}_p \mathbf{e}_1 = \mathbf{0}$  and  $\mathbf{D}_p \mathbf{e}_k = (k-1)\mathbf{e}_{k-1}$  for  $k \geq 2$ . Hence

$$\mathbf{D}_p \mathbf{S}_p \mathbf{e}_k = k \mathbf{e}_k \quad (k < p), \quad \mathbf{D}_p \mathbf{S}_p \mathbf{e}_p = \mathbf{0}, \quad \mathbf{S}_p \mathbf{D}_p \mathbf{e}_k = (k-1) \mathbf{e}_k \quad (1 \leq k \leq p),$$

where for  $k=1$  we used  $\mathbf{S}_p \mathbf{D}_p \mathbf{e}_1 = \mathbf{0} = 0 \mathbf{e}_1$ . Subtracting,  $[\mathbf{D}_p, \mathbf{S}_p] \mathbf{e}_k = \mathbf{e}_k$  for  $k < p$ , and  $[\mathbf{D}_p, \mathbf{S}_p] \mathbf{e}_p = -(p-1)\mathbf{e}_p = (1-p)\mathbf{e}_p = \mathbf{e}_p$ , because  $p \cdot 1 = 0$  in  $F$  (Definition 0.7.2, Lemma 0.7.4). So every column of  $[\mathbf{D}_p, \mathbf{S}_p]$  is the corresponding column of  $\mathbf{I}_p$ . For  $p=2$  this is the pair of Exercise (C1: The Heisenberg relation) (d), transposed.

*Warning.* “The identity is never a commutator” is a statement about characteristic 0. The trace test says that a commutator has trace 0, and  $\text{tr } \mathbf{I}_n = n \cdot 1$ . Over  $\mathbb{R}$  this is  $n \neq 0$ ; over  $\mathbb{F}_p$  it is 0 whenever  $p$  divides  $n$ , and then the test is silent. The example above shows that  $\mathbf{I}_p$  really is a commutator over  $\mathbb{F}_p$ . Whenever an argument ends with “ $n \cdot 1 = 0$ , contradiction”, check which fields it covers.

## 7.5.3 Trace-zero matrices with zero diagonal

We want to write a given trace-zero  $\mathbf{A}$  as  $[\mathbf{X}, \mathbf{Y}]$ . Look first at the easiest choice of  $\mathbf{X}$ , a diagonal matrix  $\mathbf{X} = \text{diag}(x_1, \dots, x_n)$ . For any  $\mathbf{Y} = (y_{ij})$ , Definition 0.9.9 gives  $(\mathbf{XY})_{ij} = x_i y_{ij}$  and  $(\mathbf{YX})_{ij} = y_{ij} x_j$ , so

$$[\mathbf{X}, \mathbf{Y}]_{ij} = (x_i - x_j) y_{ij}. \quad (*)$$

The diagonal entries of  $[\mathbf{X}, \mathbf{Y}]$  are all 0, and each off-diagonal entry can be anything, provided  $x_i \neq x_j$ . So a diagonal  $\mathbf{X}$  with distinct entries produces **exactly** the matrices with zero diagonal. A trace-zero matrix need not have zero diagonal, but by Proposition 7.5.1 (b) we may replace  $\mathbf{A}$  by any matrix similar to it. The plan is therefore:

- ① find a similarity that makes the diagonal of  $\mathbf{A}$  zero; ② solve  $(*)$  entrywise.

Step ① is the heart of the matter. We do it one diagonal entry at a time, starting with the  $(1, 1)$ -entry.

A matrix is **scalar** if it equals  $\lambda \mathbf{I}_n$  for some  $\lambda \in F$ . A scalar matrix is similar only to itself, since  $\mathbf{P}^{-1}(\lambda \mathbf{I}_n)\mathbf{P} = \lambda \mathbf{I}_n$ , so its diagonal cannot be changed. Every other matrix can at least be given a zero in the corner. This first lemma holds over every field.

### Lemma 7.5.2 — A Non-Scalar Matrix Has a Similar Matrix with Zero Corner

Let  $F$  be any field, let  $n \geq 2$ , and let  $\mathbf{A} \in M_n(F)$  be **not** scalar. Then there is an invertible  $\mathbf{P} \in M_n(F)$  such that the first column of  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  is  $\mathbf{e}_2$ . In particular

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} 0 & \mathbf{r}^\top \\ \mathbf{e}_1 & \mathbf{A}_1 \end{pmatrix}$$

for some  $\mathbf{r} \in F^{n-1}$  and  $\mathbf{A}_1 \in M_{n-1}(F)$  with  $\text{tr } \mathbf{A}_1 = \text{tr } \mathbf{A}$ , where  $\mathbf{e}_1 \in F^{n-1}$ .

*Idea.* Work backwards from the goal. The first column of  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  is the coordinate vector of  $\mathbf{A}\mathbf{p}_1$  in the basis formed by the columns  $\mathbf{p}_1, \dots, \mathbf{p}_n$  of  $\mathbf{P}$ . For it to be  $\mathbf{e}_2$ , we need  $\mathbf{A}\mathbf{p}_1 = \mathbf{p}_2$ . So we want a vector  $\mathbf{v}$  such that  $\mathbf{v}$  and  $\mathbf{A}\mathbf{v}$  are independent, and then we build a basis starting  $\mathbf{v}, \mathbf{A}\mathbf{v}$ . Such a  $\mathbf{v}$  exists unless  $\mathbf{A}$  sends every vector to a multiple of itself, and a short argument shows that this forces  $\mathbf{A}$  to be scalar.

*Proof.* We first show that there is  $\mathbf{v} \in F^n$  with  $(\mathbf{v}, \mathbf{A}\mathbf{v})$  linearly independent. Suppose not. Then for every  $\mathbf{v} \neq \mathbf{0}$  there are  $a, b \in F$ , not both zero, with  $a\mathbf{v} + b\mathbf{A}\mathbf{v} = \mathbf{0}$ . If  $b = 0$ , then  $a \neq 0$  and  $a\mathbf{v} = \mathbf{0}$ , which forces  $\mathbf{v} = \mathbf{0}$  (Theorem 1.1.9), a contradiction; so  $b \neq 0$  and  $\mathbf{A}\mathbf{v} = \lambda_{\mathbf{v}}\mathbf{v}$  with  $\lambda_{\mathbf{v}} = -a/b$ . Apply this to  $\mathbf{e}_i, \mathbf{e}_j$  and  $\mathbf{e}_i + \mathbf{e}_j$  for  $i \neq j$ :

$$\lambda_{\mathbf{e}_i+\mathbf{e}_j}\mathbf{e}_i + \lambda_{\mathbf{e}_i+\mathbf{e}_j}\mathbf{e}_j = \mathbf{A}(\mathbf{e}_i + \mathbf{e}_j) = \mathbf{A}\mathbf{e}_i + \mathbf{A}\mathbf{e}_j = \lambda_{\mathbf{e}_i}\mathbf{e}_i + \lambda_{\mathbf{e}_j}\mathbf{e}_j.$$

Since  $\mathbf{e}_i, \mathbf{e}_j$  are linearly independent, comparing coefficients (Theorem 1.4.3) gives  $\lambda_{\mathbf{e}_i} = \lambda_{\mathbf{e}_i+\mathbf{e}_j} = \lambda_{\mathbf{e}_j}$ . Hence all  $\lambda_{\mathbf{e}_i}$  equal a common  $\lambda$ , so  $\mathbf{A}\mathbf{e}_i = \lambda\mathbf{e}_i$  for every  $i$ , and by Theorem 0.9.2 the columns of  $\mathbf{A}$  are those of  $\lambda\mathbf{I}_n$ . So  $\mathbf{A} = \lambda\mathbf{I}_n$  is scalar, contrary to the hypothesis.

Take such a  $\mathbf{v}$ . By Theorem 1.6.5, the independent list  $(\mathbf{v}, \mathbf{A}\mathbf{v})$  extends to a basis  $(\mathbf{v}, \mathbf{A}\mathbf{v}, \mathbf{w}_3, \dots, \mathbf{w}_n)$  of  $F^n$ . Let  $\mathbf{P}$  be the matrix with these columns; it is invertible by Theorem 2.4.4. Then  $\mathbf{P}\mathbf{e}_1 = \mathbf{v}$  and  $\mathbf{P}\mathbf{e}_2 = \mathbf{A}\mathbf{v}$  (Theorem 0.9.3), so the first column of  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  is

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P}\mathbf{e}_1 = \mathbf{P}^{-1}\mathbf{A}\mathbf{v} = \mathbf{P}^{-1}\mathbf{P}\mathbf{e}_2 = \mathbf{e}_2.$$

Partitioning  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  after the first row and column gives the displayed form. Finally,  $\text{tr}(\mathbf{P}^{-1}\mathbf{A}\mathbf{P}) = \text{tr } \mathbf{A}$  by Theorem 3.7.5, and  $\text{tr}(\mathbf{P}^{-1}\mathbf{A}\mathbf{P}) = 0 + \text{tr } \mathbf{A}_1$  by Definition 0.9.12, so  $\text{tr } \mathbf{A}_1 = \text{tr } \mathbf{A}$ . ■

*Quick check.* Over  $\mathbb{Q}$ , let  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & -2 \end{pmatrix}$ . Using  $\mathbf{v} = \mathbf{e}_2$  in the proof of Lemma 7.5.2, find  $\mathbf{P}$  and  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ , and check that its diagonal is zero.

*Solution.*  $\mathbf{A}\mathbf{e}_2 = (1, -2)$ , which is not a multiple of  $\mathbf{e}_2$ , so  $(\mathbf{e}_2, \mathbf{A}\mathbf{e}_2)$  is independent and already a basis of  $\mathbb{Q}^2$ . Then  $\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix}$ , with  $\det \mathbf{P} = -1$  and  $\mathbf{P}^{-1} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$ . We

get  $\mathbf{AP} = \begin{pmatrix} 1 & 0 \\ -2 & 4 \end{pmatrix}$  and  $\mathbf{P}^{-1}\mathbf{AP} = \begin{pmatrix} 0 & 4 \\ 1 & 0 \end{pmatrix}$ . The first column is  $\mathbf{e}_2$ , as the lemma promises; the  $(2, 2)$ -entry is 0 as well, because the trace is 0 and trace is a similarity invariant.

The lemma controls one diagonal entry and hands us a smaller matrix  $\mathbf{A}_1$  with the same trace. That is exactly the setting for induction on the size. The only thing that can stop the induction is a scalar  $\mathbf{A}_1$ , and here the trace saves us, provided the field has characteristic 0.

### Lemma 7.5.3 — Trace Zero Means Similar to Zero Diagonal

Let  $F$  be a field of **characteristic** 0, let  $n \geq 1$ , and let  $\mathbf{A} \in M_n(F)$  with  $\text{tr } \mathbf{A} = 0$ . Then  $\mathbf{A}$  is similar to a matrix all of whose diagonal entries are 0.

*Idea.* Induction on  $n$ . If  $\mathbf{A}$  is scalar,  $\mathbf{A} = \lambda \mathbf{I}_n$  with  $n\lambda = 0$ , and characteristic 0 forces  $\lambda = 0$ : then  $\mathbf{A} = 0$  already has zero diagonal. Otherwise, Lemma 7.5.2 puts a 0 in the corner and leaves a trace-zero block  $\mathbf{A}_1$  of size  $n-1$ . The induction hypothesis conjugates  $\mathbf{A}_1$  by some  $\mathbf{Q}$  into zero diagonal. Conjugating the whole matrix by the block diagonal matrix  $(1) \oplus \mathbf{Q}$  does this to the lower block without disturbing the corner.

*Proof.* We use induction on  $n$  (Theorem 0.2.7). For  $n = 1$ ,  $\mathbf{A} = (\text{tr } \mathbf{A}) = (0)$ , which has zero diagonal.

Let  $n \geq 2$ , and assume the statement for trace-zero matrices of size  $n-1$ .

*Case 1:  $\mathbf{A}$  is scalar,* say  $\mathbf{A} = \lambda \mathbf{I}_n$ . Then  $0 = \text{tr } \mathbf{A} = n \cdot \lambda = (n \cdot 1)\lambda$  by Definition 0.9.12 and Lemma 0.7.4. Since  $F$  has characteristic 0 and  $n \geq 1$ ,  $n \cdot 1 \neq 0$  (Definition 0.7.2), so  $\lambda = 0$ . Hence  $\mathbf{A} = 0$ , which has zero diagonal and is similar to itself.

*Case 2:  $\mathbf{A}$  is not scalar.* By Lemma 7.5.2 there is an invertible  $\mathbf{P}$  with

$$\mathbf{M} := \mathbf{P}^{-1}\mathbf{AP} = \begin{pmatrix} 0 & \mathbf{r}^\top \\ \mathbf{e}_1 & \mathbf{A}_1 \end{pmatrix}, \quad \text{tr } \mathbf{A}_1 = \text{tr } \mathbf{A} = 0.$$

By the induction hypothesis there is an invertible  $\mathbf{Q} \in M_{n-1}(F)$  such that  $\mathbf{Z}_1 := \mathbf{Q}^{-1}\mathbf{A}_1\mathbf{Q}$  has zero diagonal. Let  $\mathbf{R} = (1) \oplus \mathbf{Q} = \begin{pmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{Q} \end{pmatrix}$ . By Theorem 7.1.4,  $\mathbf{R}$  is invertible with  $\mathbf{R}^{-1} = (1) \oplus \mathbf{Q}^{-1}$ . By Theorem 7.1.1,

$$\begin{aligned} \mathbf{R}^{-1}\mathbf{MR} &= \begin{pmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{Q}^{-1} \end{pmatrix} \begin{pmatrix} 0 & \mathbf{r}^\top \\ \mathbf{e}_1 & \mathbf{A}_1 \end{pmatrix} \begin{pmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{Q} \end{pmatrix} \\ &= \begin{pmatrix} 0 & \mathbf{r}^\top \mathbf{Q} \\ \mathbf{Q}^{-1}\mathbf{e}_1 & \mathbf{Q}^{-1}\mathbf{A}_1\mathbf{Q} \end{pmatrix} = \begin{pmatrix} 0 & \mathbf{r}^\top \mathbf{Q} \\ \mathbf{Q}^{-1}\mathbf{e}_1 & \mathbf{Z}_1 \end{pmatrix}. \end{aligned}$$

Its  $(1, 1)$ -entry is 0, and its other diagonal entries are those of  $\mathbf{Z}_1$ , which are 0. Since  $\mathbf{R}^{-1}\mathbf{MR} = (\mathbf{PR})^{-1}\mathbf{A}(\mathbf{PR})$  by Theorem 0.9.7 (3), and  $\mathbf{PR}$  is invertible,  $\mathbf{A}$  is similar to a matrix with zero diagonal. This completes the induction. ■

*Remark.* The characteristic was used only in Case 1, and only through  $k \cdot 1 \neq 0$  for the sizes  $k = 2, \dots, n$  that the induction passes through. So the lemma holds, with the same

proof, whenever  $F$  has characteristic 0 or characteristic  $p > n$ . Some hypothesis on the characteristic is needed: for  $n = p$ , over  $\mathbb{F}_p$ , the matrix  $\mathbf{I}_p$  has trace  $p \cdot 1 = 0$ , but it is scalar, so it is similar only to itself, and its diagonal entries are 1. The same happens for  $\lambda \mathbf{I}_n$  with  $\lambda \neq 0$  whenever  $p$  divides  $n$ .

## 7.5.4 Shoda's theorem

With a zero diagonal available, the diagonal choice of  $\mathbf{X}$  in (\*) finishes the job. All we need are  $n$  distinct scalars for the diagonal of  $\mathbf{X}$ , and characteristic 0 provides them.

### Theorem 7.5.4 — Shoda's Theorem

Let  $F$  be a field of **characteristic** 0, let  $n \geq 1$ , and let  $\mathbf{A} \in M_n(F)$ . Then  $\mathbf{A}$  is a commutator, that is  $\mathbf{A} = [\mathbf{X}, \mathbf{Y}]$  for some  $\mathbf{X}, \mathbf{Y} \in M_n(F)$ , if and only if  $\text{tr } \mathbf{A} = 0$ .

*Idea.* One direction is Proposition 7.5.1 (c). For the other: ① by Lemma 7.5.3,  $\mathbf{A} = \mathbf{PZP}^{-1}$  with  $\mathbf{Z}$  of zero diagonal; ② take  $\mathbf{X}_0 = \text{diag}(0, 1, \dots, n-1)$ , whose entries are distinct in characteristic 0; ③ solve  $[\mathbf{X}_0, \mathbf{Y}_0] = \mathbf{Z}$  entrywise with (\*), dividing by  $x_i - x_j \neq 0$ ; ④ conjugate back with Proposition 7.5.1 (b).

*Proof.* ( $\Rightarrow$ ) If  $\mathbf{A} = [\mathbf{X}, \mathbf{Y}]$ , then  $\text{tr } \mathbf{A} = 0$  by Proposition 7.5.1 (c).

( $\Leftarrow$ ) Suppose  $\text{tr } \mathbf{A} = 0$ . By Lemma 7.5.3 there is an invertible  $\mathbf{P}$  such that  $\mathbf{Z} = (z_{ij}) := \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  has  $z_{ii} = 0$  for all  $i$ .

For  $1 \leq i \leq n$  let  $x_i := (i-1) \cdot 1 \in F$ , and  $\mathbf{X}_0 := \text{diag}(x_1, \dots, x_n)$ . For  $i < j$ , Lemma 0.7.4 gives  $x_j - x_i = (j-i) \cdot 1$ , which is non-zero because  $j-i \geq 1$  and  $F$  has characteristic 0 (Definition 0.7.2). So the  $x_i$  are distinct, and we may define  $\mathbf{Y}_0 = (y_{ij}) \in M_n(F)$  by

$$y_{ij} := \frac{z_{ij}}{x_i - x_j} \quad (i \neq j), \quad y_{ii} := 0.$$

By (\*),  $[\mathbf{X}_0, \mathbf{Y}_0]_{ij} = (x_i - x_j)y_{ij} = z_{ij}$  for  $i \neq j$ , and  $[\mathbf{X}_0, \mathbf{Y}_0]_{ii} = 0 = z_{ii}$ . Hence  $[\mathbf{X}_0, \mathbf{Y}_0] = \mathbf{Z}$ .

Let  $\mathbf{X} = \mathbf{P}\mathbf{X}_0\mathbf{P}^{-1}$  and  $\mathbf{Y} = \mathbf{P}\mathbf{Y}_0\mathbf{P}^{-1}$ . Applying Proposition 7.5.1 (b) with the invertible matrix  $\mathbf{P}^{-1}$ ,

$$[\mathbf{X}, \mathbf{Y}] = [\mathbf{P}\mathbf{X}_0\mathbf{P}^{-1}, \mathbf{P}\mathbf{Y}_0\mathbf{P}^{-1}] = \mathbf{P}[\mathbf{X}_0, \mathbf{Y}_0]\mathbf{P}^{-1} = \mathbf{PZP}^{-1} = \mathbf{A}.$$

This proves that  $\mathbf{A}$  is a commutator. ■

The proof used characteristic 0 twice: in the lemma, to rule out non-zero scalar blocks of trace zero, and here, to find  $n$  distinct diagonal entries. The second use needs only  $(j-i) \cdot 1 \neq 0$  for  $1 \leq j-i \leq n-1$ , and with the remark after Lemma 7.5.3, the same proof works whenever the characteristic of  $F$  is 0 or larger than  $n$ . Neither use is essential for the **theorem**. In fact the statement “a square matrix over a field is a commutator if and only if its trace is 0” is true over **every** field. The proof in positive characteristic needs a different normal form and more work, and we do not give it. Example (The Identity as a Commutator in Characteristic  $p$ ) is a small

instance:  $\mathbf{I}_p$  over  $\mathbb{F}_p$  is a commutator, although [Lemma 7.5.3](#) fails for it.

The proof is constructive, and it is worth running once in full.

**Example. (Writing a Trace-Zero  $3 \times 3$  Matrix as a Commutator)**

Over  $\mathbb{Q}$ , write  $\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & -2 \end{pmatrix}$  as a commutator  $[\mathbf{X}, \mathbf{Y}]$ .

*Solution.* The trace is  $1 + 1 - 2 = 0$ . We follow the proofs step by step.

*First corner.*  $\mathbf{A}$  is not scalar. Take  $\mathbf{v} = \mathbf{e}_1$ ; then  $\mathbf{A}\mathbf{v} = (1, 0, 1)$  is not a multiple of  $\mathbf{e}_1$ . Extend  $(\mathbf{e}_1, \mathbf{A}\mathbf{e}_1)$  by  $\mathbf{e}_2$ , which is not in their span, to a basis, and let

$$\mathbf{P}_1 = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{M} = \mathbf{P}_1^{-1} \mathbf{A} \mathbf{P}_1 = \begin{pmatrix} 0 & 2 & 2 \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

(To check  $\mathbf{M}$  without inverting, verify  $\mathbf{P}_1 \mathbf{M} = \mathbf{A} \mathbf{P}_1$ : both equal  $\begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 1 & -1 & 0 \end{pmatrix}$ .) The

first column is  $\mathbf{e}_2$ , and the lower block is  $\mathbf{A}_1 = \begin{pmatrix} -1 & 0 \\ 1 & 1 \end{pmatrix}$ , with trace 0.

*Second corner.*  $\mathbf{A}_1$  is not scalar. With  $\mathbf{w} = \mathbf{e}_1 \in \mathbb{Q}^2$ ,  $\mathbf{A}_1 \mathbf{w} = (-1, 1)$  is independent of  $\mathbf{e}_1$ , so

$$\mathbf{Q} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{Q}^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{Z}_1 = \mathbf{Q}^{-1} \mathbf{A}_1 \mathbf{Q} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

*Assemble.* With  $\mathbf{r}^\top = (2, 2)$ , the proof of [Lemma 7.5.3](#) gives  $\mathbf{r}^\top \mathbf{Q} = (2, 0)$  and  $\mathbf{Q}^{-1} \mathbf{e}_1 = \mathbf{e}_1$ , so for  $\mathbf{P} = \mathbf{P}_1((1) \oplus \mathbf{Q})$ ,

$$\mathbf{P} = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix}, \quad \mathbf{Z} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \begin{pmatrix} 0 & 2 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{P}^{-1} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

*Solve entrywise.* With  $\mathbf{X}_0 = \text{diag}(0, 1, 2)$ , the rule  $y_{ij} = z_{ij}/(x_i - x_j)$  gives  $y_{12} = \frac{2}{-1} = -2$ ,  $y_{21} = \frac{1}{1} = 1$ ,  $y_{23} = \frac{1}{-1} = -1$ ,  $y_{32} = \frac{1}{1} = 1$ , and  $y_{13} = y_{31} = 0$ :

$$\mathbf{Y}_0 = \begin{pmatrix} 0 & -2 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}.$$

*Conjugate back.*  $\mathbf{X} = \mathbf{P} \mathbf{X}_0 \mathbf{P}^{-1}$  and  $\mathbf{Y} = \mathbf{P} \mathbf{Y}_0 \mathbf{P}^{-1}$  are

$$\mathbf{X} = \begin{pmatrix} 0 & -1 & 1 \\ 0 & 2 & 0 \\ 0 & -1 & 1 \end{pmatrix}, \quad \mathbf{Y} = \begin{pmatrix} 1 & -4 & -4 \\ 0 & 1 & 1 \\ 1 & -2 & -2 \end{pmatrix}.$$

$$\text{Check: } \mathbf{XY} = \begin{pmatrix} 1 & -3 & -3 \\ 0 & 2 & 2 \\ 1 & -3 & -3 \end{pmatrix} \text{ and } \mathbf{YX} = \begin{pmatrix} 0 & -5 & -3 \\ 0 & 1 & 1 \\ 0 & -3 & -1 \end{pmatrix}, \text{ and } \mathbf{XY} - \mathbf{YX} = \mathbf{A}.$$

Nothing about these  $\mathbf{X}$  and  $\mathbf{Y}$  is unique. Adding to  $\mathbf{Y}$  any matrix that commutes with  $\mathbf{X}$ , for instance a polynomial in  $\mathbf{X}$ , gives another solution, by [Proposition 7.5.1](#) (a).

## 7.5.5 The span of the commutators

Shoda's theorem is about single commutators. A weaker question, with an easier answer that holds over **every** field, is which matrices are **sums** of commutators. Since the trace-zero matrices form a subspace and contain all commutators, the span of the commutators lies inside it. Matrix units show that it is all of it.

### Theorem 7.5.5 — Commutators Span the Trace-Zero Matrices

Let  $F$  be any field and  $n \geq 1$ . The span of all commutators  $[\mathbf{X}, \mathbf{Y}]$  with  $\mathbf{X}, \mathbf{Y} \in M_n(F)$  is the subspace  $\{\mathbf{A} \in M_n(F) : \text{tr } \mathbf{A} = 0\}$ . More precisely, every trace-zero matrix is a linear combination of the commutators

$$\mathbf{E}_{ij} = [\mathbf{E}_{ii}, \mathbf{E}_{ij}] \quad (i \neq j) \quad \text{and} \quad \mathbf{E}_{ii} - \mathbf{E}_{nn} = [\mathbf{E}_{in}, \mathbf{E}_{ni}] \quad (1 \leq i \leq n-1).$$

*Proof.* Let  $W = \{\mathbf{A} \in M_n(F) : \text{tr } \mathbf{A} = 0\}$ , a subspace by Example (Trace-Free Matrices). By [Proposition 7.5.1](#) (c), every commutator lies in  $W$ , so their span is contained in  $W$  (Theorem 1.3.1, the span being the smallest subspace containing them).

For the reverse inclusion, the displayed identities hold by Example 2 after [Definition 7.5.1](#): for  $i \neq j$ ,  $[\mathbf{E}_{ii}, \mathbf{E}_{ij}] = \mathbf{E}_{ij}$ , and for  $i < n$ ,  $[\mathbf{E}_{in}, \mathbf{E}_{ni}] = \mathbf{E}_{ii} - \mathbf{E}_{nn}$ . Now let  $\mathbf{A} = (a_{ij}) \in W$ . Writing  $\mathbf{A}$  in the basis of matrix units (Example (Standard Bases)) and using  $a_{nn} = -\sum_{i < n} a_{ii}$ ,

$$\mathbf{A} = \sum_{i \neq j} a_{ij} \mathbf{E}_{ij} + \sum_{i=1}^{n-1} a_{ii} \mathbf{E}_{ii} + a_{nn} \mathbf{E}_{nn} = \sum_{i \neq j} a_{ij} \mathbf{E}_{ij} + \sum_{i=1}^{n-1} a_{ii} (\mathbf{E}_{ii} - \mathbf{E}_{nn}).$$

So  $\mathbf{A}$  is a linear combination of commutators, hence lies in their span. (For  $n = 1$ ,  $W = \{0\}$  and both sides are  $\{0\}$ .)

■

The theorem lists  $n(n-1) + (n-1) = n^2 - 1$  commutators spanning the trace-zero matrices. Since  $\text{tr}: M_n(F) \rightarrow F$  is onto ( $\text{tr}(c\mathbf{E}_{11}) = c$ ), the Rank–Nullity Theorem (Theorem 3.3.1) gives that its kernel has dimension  $n^2 - 1$ , so by Theorem 1.6.4 (b) these commutators form a basis of it. Over a field of characteristic 0, Shoda's theorem says much more: the sum of commutators produced here can always be collapsed into a single commutator. In general the set of commutators is not obviously closed under addition, and that is exactly what makes Shoda's theorem a theorem.

**Warning.** The span of a set is not the set. [Theorem 7.5.5](#) shows that every trace-zero matrix is a **sum** of commutators, over any field, by a two-line argument. It does not show that every trace-zero matrix **is** a commutator. For comparison, in  $M_2(F)$  the

matrices of rank at most 1 span everything (the four matrix units have rank 1), yet  $\mathbf{I}_2$  has rank 2: a set can span a subspace without filling it. For commutators it happens that the set does fill the subspace, but the proof needs the similarity argument of [Lemma 7.5.3](#), not just matrix units.

The span theorem gives a clean proof of a fact from Chapter 3: a linear functional  $\varphi$  on  $M_n(F)$  with  $\varphi(\mathbf{AB}) = \varphi(\mathbf{BA})$  is a multiple of the trace. Such a  $\varphi$  kills every commutator, hence their span, which is the kernel of the trace; [Exercise \(C1: Trace-like functionals, again\)](#) below completes the argument.

## 7.5.6 Exercises

### 7.5.6.1 A. Check your understanding

#### Exercise A1.

- Define the commutator  $[\mathbf{A}, \mathbf{B}]$ , and state what it means for a matrix to be a commutator.
- State Shoda's theorem, including the hypothesis on the field.
- True or false:  $[\mathbf{A}, \mathbf{B}] = [\mathbf{B}, \mathbf{A}]$  for all  $\mathbf{A}, \mathbf{B} \in M_n(F)$ . Justify your answer.
- True or false: over every field, every trace-zero matrix is similar to a matrix with zero diagonal. Justify your answer.
- Name the two places where the proof of Shoda's theorem uses characteristic 0.
- Explain why [Theorem 7.5.5](#) does not by itself prove Shoda's theorem.

*Solution.*

- $[\mathbf{A}, \mathbf{B}] = \mathbf{AB} - \mathbf{BA}$  ([Definition 7.5.1](#)). A matrix  $\mathbf{C} \in M_n(F)$  is a commutator if  $\mathbf{C} = [\mathbf{X}, \mathbf{Y}]$  for some  $\mathbf{X}, \mathbf{Y} \in M_n(F)$ .
- If  $F$  has characteristic 0 and  $\mathbf{A} \in M_n(F)$ , then  $\mathbf{A} = [\mathbf{X}, \mathbf{Y}]$  for some  $\mathbf{X}, \mathbf{Y} \in M_n(F)$  if and only if  $\text{tr } \mathbf{A} = 0$  ([Theorem 7.5.4](#)).
- False.  $[\mathbf{B}, \mathbf{A}] = -[\mathbf{A}, \mathbf{B}]$  ([Proposition 7.5.1](#) (a)), so equality holds only when  $[\mathbf{A}, \mathbf{B}] = 0$ . For example  $[\mathbf{E}_{12}, \mathbf{E}_{21}] = \text{diag}(1, -1)$  and  $[\mathbf{E}_{21}, \mathbf{E}_{12}] = \text{diag}(-1, 1)$  over  $\mathbb{Q}$ .
- False. Over  $\mathbb{F}_2$ ,  $\mathbf{I}_2$  has trace  $1 + 1 = 0$ , but it is scalar, so every matrix similar to it equals  $\mathbf{P}^{-1}\mathbf{I}_2\mathbf{P} = \mathbf{I}_2$ , whose diagonal entries are 1.
- In [Lemma 7.5.3](#), to conclude from  $(n \cdot 1)\lambda = 0$  that a trace-zero scalar matrix  $\lambda\mathbf{I}_n$  is 0; and in [Theorem 7.5.4](#), to find  $n$  distinct scalars  $x_i = (i - 1) \cdot 1$  so that we can divide by  $x_i - x_j$ .
- It says every trace-zero matrix is a **linear combination** of commutators. Shoda's theorem says it is a **single** commutator. A sum of commutators need not visibly be a commutator, since  $[\mathbf{X}, \mathbf{Y}] + [\mathbf{X}', \mathbf{Y}']$  has no evident expression as  $[\mathbf{X}'', \mathbf{Y}']$ .

### 7.5.6.2 B. Practice

**Exercise B1: A  $2 \times 2$  commutator.**

Over  $\mathbb{Q}$ , let  $\mathbf{A} = \begin{pmatrix} 3 & -2 \\ 1 & -3 \end{pmatrix}$ .

- (a) Using  $\mathbf{v} = \mathbf{e}_1$  as in [Lemma 7.5.2](#), find an invertible  $\mathbf{P}$  such that  $\mathbf{Z} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  has zero diagonal.
- (b) With  $\mathbf{X}_0 = \text{diag}(0, 1)$ , find  $\mathbf{Y}_0$  with  $[\mathbf{X}_0, \mathbf{Y}_0] = \mathbf{Z}$ .
- (c) Hence write  $\mathbf{A} = [\mathbf{X}, \mathbf{Y}]$ , and check your answer by multiplying out.

*Solution.*

- (a)  $\text{tr } \mathbf{A} = 0$ , and  $\mathbf{A}\mathbf{e}_1 = (3, 1)$  is not a multiple of  $\mathbf{e}_1$ , so  $(\mathbf{e}_1, \mathbf{A}\mathbf{e}_1)$  is a basis of  $\mathbb{Q}^2$ .

Let  $\mathbf{P} = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$ , so  $\mathbf{P}^{-1} = \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix}$ . Then  $\mathbf{A}\mathbf{P} = \begin{pmatrix} 3 & 7 \\ 1 & 0 \end{pmatrix}$  and

$$\mathbf{Z} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} 0 & 7 \\ 1 & 0 \end{pmatrix}.$$

The first column is  $\mathbf{e}_2$  by [Lemma 7.5.2](#), and the  $(2, 2)$ -entry is 0 because the trace is 0.

- (b) By (\*),  $[\mathbf{X}_0, \mathbf{Y}]_{12} = (0 - 1)y_{12}$  and  $[\mathbf{X}_0, \mathbf{Y}]_{21} = (1 - 0)y_{21}$ , while the diagonal of  $[\mathbf{X}_0, \mathbf{Y}]$  is zero. So  $y_{12} = -7$ ,  $y_{21} = 1$ , and we may take  $y_{11} = y_{22} = 0$ :

$$\mathbf{Y}_0 = \begin{pmatrix} 0 & -7 \\ 1 & 0 \end{pmatrix}.$$

- (c) By the proof of [Theorem 7.5.4](#),  $\mathbf{X} = \mathbf{P}\mathbf{X}_0\mathbf{P}^{-1}$  and  $\mathbf{Y} = \mathbf{P}\mathbf{Y}_0\mathbf{P}^{-1}$ :

$$\mathbf{X} = \begin{pmatrix} 0 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 3 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{Y} = \begin{pmatrix} 3 & -7 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 3 & -16 \\ 1 & -3 \end{pmatrix},$$

using  $\mathbf{P}\mathbf{X}_0 = \begin{pmatrix} 0 & 3 \\ 0 & 1 \end{pmatrix}$  and  $\mathbf{P}\mathbf{Y}_0 = \begin{pmatrix} 3 & -7 \\ 1 & 0 \end{pmatrix}$ . Check:  $\mathbf{X}\mathbf{Y} = \begin{pmatrix} 3 & -9 \\ 1 & -3 \end{pmatrix}$ ,  $\mathbf{Y}\mathbf{X} = \begin{pmatrix} 0 & -7 \\ 0 & 0 \end{pmatrix}$ , and  $\mathbf{X}\mathbf{Y} - \mathbf{Y}\mathbf{X} = \begin{pmatrix} 3 & -2 \\ 1 & -3 \end{pmatrix} = \mathbf{A}$ .

**Exercise B2: A commutator over  $\mathbb{F}_2$ .**

Work over  $\mathbb{F}_2$ , and let  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ .

- (a) Show that  $\text{tr } \mathbf{A} = 0$  and that  $\mathbf{A}$  is not scalar.
- (b) Explain why  $\mathbf{v} = \mathbf{e}_1$  cannot be used in [Lemma 7.5.2](#), and use  $\mathbf{v} = \mathbf{e}_2$  instead to find  $\mathbf{P}$  with  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  of zero diagonal.
- (c) The field  $\mathbb{F}_2$  has exactly two elements. Use them as the diagonal of  $\mathbf{X}_0$  to write  $\mathbf{A}$  as a commutator, by hand.
- (d) Explain why the same method cannot write  $\mathbf{I}_2 \in M_2(\mathbb{F}_2)$  as a commutator, and recall a pair  $\mathbf{X}, \mathbf{Y}$  that does.

*Solution.*

- (a)  $\text{tr } \mathbf{A} = 1 + 1 = 0$  in  $\mathbb{F}_2$ .  $\mathbf{A}$  has a non-zero off-diagonal entry, while every scalar matrix has zero off-diagonal entries.
- (b)  $\mathbf{A}\mathbf{e}_1 = \mathbf{e}_1$ , so  $(\mathbf{e}_1, \mathbf{A}\mathbf{e}_1)$  is dependent. With  $\mathbf{v} = \mathbf{e}_2$ ,  $\mathbf{A}\mathbf{e}_2 = (1, 1)$ , which is not a multiple of  $\mathbf{e}_2$ , so  $(\mathbf{e}_2, \mathbf{A}\mathbf{e}_2)$  is a basis. Let  $\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ ; since  $\det \mathbf{P} = -1 = 1$ ,  $\mathbf{P}^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$  (check:  $\mathbf{P}\mathbf{P}^{-1} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = \mathbf{I}_2$  in  $\mathbb{F}_2$ ). Then  $\mathbf{A}\mathbf{P} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$  and  $\mathbf{Z} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ .
- (c) Take  $\mathbf{X}_0 = \text{diag}(0, 1)$ . By (\*),  $[\mathbf{X}_0, \mathbf{Y}]_{12} = (0 - 1)y_{12} = y_{12}$  (as  $-1 = 1$ ) and  $[\mathbf{X}_0, \mathbf{Y}]_{21} = y_{21}$ . So  $\mathbf{Y}_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  gives  $[\mathbf{X}_0, \mathbf{Y}_0] = \mathbf{Z}$ . Conjugating back,  $\mathbf{P}\mathbf{X}_0 = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$  and  $\mathbf{P}\mathbf{Y}_0 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ , so

$$\mathbf{X} = \mathbf{P}\mathbf{X}_0\mathbf{P}^{-1} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{Y} = \mathbf{P}\mathbf{Y}_0\mathbf{P}^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Check:  $\mathbf{X}\mathbf{Y} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ ,  $\mathbf{Y}\mathbf{X} = \begin{pmatrix} 2 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ , so  $[\mathbf{X}, \mathbf{Y}] = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \mathbf{A}$ .  
(Here  $\mathbf{Y} = \mathbf{A}$ : indeed  $[\mathbf{X}, \mathbf{A}] = \mathbf{A}$ .)

- (d) The method writes  $\mathbf{Z} = \mathbf{P}^{-1}\mathbf{I}_2\mathbf{P}$  as  $[\mathbf{X}_0, \mathbf{Y}_0]$  with  $\mathbf{X}_0$  diagonal, and by (\*) such a commutator has zero diagonal. But  $\mathbf{I}_2$  is scalar, so  $\mathbf{P}^{-1}\mathbf{I}_2\mathbf{P} = \mathbf{I}_2$  for every invertible  $\mathbf{P}$ , and its diagonal entries are 1. So the method cannot start. Nevertheless  $\mathbf{I}_2$  is a commutator: with  $\mathbf{X} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$  and  $\mathbf{Y} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ ,  $[\mathbf{X}, \mathbf{Y}] = \text{diag}(-1, 1) = \mathbf{I}_2$  in  $\mathbb{F}_2$  (Exercise (C1: The Heisenberg relation) (d), or [Example \(The Identity as a Commutator in Characteristic  \$p\$ \)](#) with  $p = 2$ ).

### Exercise B3: Commuting with a diagonal matrix.

Let  $F$  be any field, let  $d_1, \dots, d_n \in F$  be **distinct**, let  $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$ , and define  $L: M_n(F) \rightarrow M_n(F)$  by  $L(\mathbf{X}) = [\mathbf{D}, \mathbf{X}]$ .

- (a) Show that  $L$  is linear.
- (b) Prove that  $\ker L$  is the set of diagonal matrices and  $\text{im } L$  is the set of matrices with zero diagonal.
- (c) Check that your answer to (b) agrees with the Rank–Nullity Theorem.
- (d) Hence decide whether  $\begin{pmatrix} 0 & 1 & 2 \\ 3 & 0 & 4 \\ 5 & 6 & 0 \end{pmatrix} \in M_3(\mathbb{Q})$  is a commutator with  $\mathbf{X} = \text{diag}(1, 2, 3)$ , and if so find  $\mathbf{Y}$ .

*Solution.*

- (a) By [Proposition 7.5.1](#) (a),  $[\mathbf{D}, \mathbf{X} + \mathbf{X}'] = [\mathbf{D}, \mathbf{X}] + [\mathbf{D}, \mathbf{X}']$  and  $[\mathbf{D}, c\mathbf{X}] = c[\mathbf{D}, \mathbf{X}]$ .
- (b) By (\*),  $L(\mathbf{X})_{ij} = (d_i - d_j)x_{ij}$ . Since  $d_i - d_j \neq 0$  for  $i \neq j$ ,  $L(\mathbf{X}) = 0$  if and only

if  $x_{ij} = 0$  for all  $i \neq j$ , that is,  $\mathbf{X}$  is diagonal. Every  $L(\mathbf{X})$  has zero diagonal. Conversely, if  $\mathbf{Z}$  has zero diagonal, the matrix  $\mathbf{Y}$  with  $y_{ij} = z_{ij}/(d_i - d_j)$  for  $i \neq j$  and  $y_{ii} = 0$  satisfies  $L(\mathbf{Y}) = \mathbf{Z}$ . So  $\text{im } L$  is exactly the set of zero-diagonal matrices.

- (c) The diagonal matrices have basis  $\mathbf{E}_{11}, \dots, \mathbf{E}_{nn}$ , so nullity  $L = n$ . The zero-diagonal matrices have basis  $\mathbf{E}_{ij}$  ( $i \neq j$ ), so  $\text{rank } L = n^2 - n$ . Their sum is  $n^2 = \dim M_n(F)$ , as Theorem 3.3.1 requires.
- (d) Yes, since the matrix has zero diagonal and 1, 2, 3 are distinct. With  $d = (1, 2, 3)$ ,  $y_{ij} = z_{ij}/(d_i - d_j)$  gives

$$\mathbf{Y} = \begin{pmatrix} 0 & \frac{1}{-1} & \frac{2}{-2} \\ \frac{3}{1} & 0 & \frac{4}{-1} \\ \frac{5}{2} & \frac{6}{1} & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 & -1 \\ 3 & 0 & -4 \\ \frac{5}{2} & 6 & 0 \end{pmatrix},$$

and by (b),  $[\text{diag}(1, 2, 3), \mathbf{Y}]$  is the given matrix.

### 7.5.6.3 C. Going deeper

#### Exercise C1: Trace-like functionals, again.

Let  $F$  be any field,  $n \geq 1$ , and let  $\varphi: M_n(F) \rightarrow F$  be linear with  $\varphi(\mathbf{AB}) = \varphi(\mathbf{BA})$  for all  $\mathbf{A}, \mathbf{B} \in M_n(F)$ .

- (a) Prove that  $\varphi$  vanishes on every trace-zero matrix.
- (b) Deduce that  $\varphi = c \text{ tr}$ , where  $c = \varphi(\mathbf{E}_{11})$ .
- (c) Suppose  $F$  has characteristic 0 and, in addition,  $\varphi(\mathbf{I}_n) = n \cdot 1$ . Deduce that  $\varphi = \text{tr}$ .
- (d) Show that the conclusion of (c) fails over  $\mathbb{F}_p$  when  $n = p$ .

*Hint: for (a), use Theorem 7.5.5.*

*Solution.*

- (a) For all  $\mathbf{X}, \mathbf{Y}$ ,  $\varphi([\mathbf{X}, \mathbf{Y}]) = \varphi(\mathbf{XY}) - \varphi(\mathbf{YX}) = 0$  by linearity and the hypothesis. So the kernel of  $\varphi$ , a subspace, contains every commutator, hence their span (Theorem 1.3.1), which is the set of trace-zero matrices by Theorem 7.5.5.
- (b) Let  $\mathbf{A} \in M_n(F)$ . Since  $\text{tr } \mathbf{E}_{11} = 1$ , the matrix  $\mathbf{A} - (\text{tr } \mathbf{A})\mathbf{E}_{11}$  has trace 0 (Theorem 0.9.6 (1)), so by (a) and linearity,

$$0 = \varphi(\mathbf{A} - (\text{tr } \mathbf{A})\mathbf{E}_{11}) = \varphi(\mathbf{A}) - (\text{tr } \mathbf{A})\varphi(\mathbf{E}_{11}).$$

Hence  $\varphi(\mathbf{A}) = c \text{ tr } \mathbf{A}$  with  $c = \varphi(\mathbf{E}_{11})$ . This recovers Exercise (C2: The trace is the only trace-like functional) by a different route: the matrix-unit computations are now packaged in Theorem 7.5.5.

- (c) By (b),  $n \cdot 1 = \varphi(\mathbf{I}_n) = c \text{ tr } \mathbf{I}_n = c(n \cdot 1)$ . Since  $F$  has characteristic 0,  $n \cdot 1 \neq 0$ , so  $c = 1$  and  $\varphi = \text{tr}$ .
- (d) Over  $\mathbb{F}_p$  with  $n = p$ , take  $\varphi = 0$ . It is linear, satisfies  $\varphi(\mathbf{AB}) = 0 = \varphi(\mathbf{BA})$ , and  $\varphi(\mathbf{I}_p) = 0 = p \cdot 1$ . But  $\varphi \neq \text{tr}$ , since  $\text{tr } \mathbf{E}_{11} = 1$ . The step that fails is canceling  $n \cdot 1 = 0$  in (c).

**Exercise C2: When  $\mathbf{A}$  commutes with  $[\mathbf{A}, \mathbf{B}]$ .**

Let  $F$  be a field of characteristic 0, let  $\mathbf{A}, \mathbf{B} \in M_n(F)$ , put  $\mathbf{C} = [\mathbf{A}, \mathbf{B}]$ , and suppose that  $\mathbf{A}\mathbf{C} = \mathbf{C}\mathbf{A}$ . The goal is to prove that  $\mathbf{C}$  is **nilpotent**, that is,  $\mathbf{C}^s = 0$  for some  $s \geq 1$ .

- (a) Prove that  $\mathbf{C}^k = [\mathbf{A}, \mathbf{B}\mathbf{C}^{k-1}]$  for every  $k \geq 1$ . Deduce that  $\text{tr } \mathbf{C}^k = 0$  for every  $k \geq 1$ .
- (b) Show that there are an integer  $s \geq 1$  and  $g \in F[x]$  with  $g(0) \neq 0$  such that  $\mathbf{C}^s g(\mathbf{C}) = 0$ .
- (c) Let  $U = \text{col}(\mathbf{C}^s)$  and  $r = \dim U$ , and suppose  $r \geq 1$ . Extend a basis of  $U$  to a basis of  $F^n$ , and let  $\mathbf{P}$  be the matrix with these columns. Show that

$$\mathbf{P}^{-1}\mathbf{C}\mathbf{P} = \begin{pmatrix} \mathbf{D} & * \\ 0 & \mathbf{N} \end{pmatrix}, \quad \mathbf{D} \in M_r(F),$$

with  $\mathbf{N}^s = 0$  and  $g(\mathbf{D}) = 0$ .

- (d) Write  $g = g_0 + xh$  with  $g_0 = g(0) \neq 0$ . Show that  $\mathbf{I}_r$  is a linear combination of  $\mathbf{D}^s, \mathbf{D}^{s+1}, \dots, \mathbf{D}^d$  for some  $d \geq s$ , and use (a) to reach a contradiction.
- (e) Conclude that  $\mathbf{C}^s = 0$ .

*Hint: in (c), compare the columns of  $\mathbf{P}^{-1}\mathbf{C}^k\mathbf{P}$  for  $k = 1$  and  $k = s$  with the coordinates of vectors of  $U$ ; in (d), take traces.*

*Solution.*

- (a) First,  $\mathbf{C}^{k-1}\mathbf{A} = \mathbf{A}\mathbf{C}^{k-1}$  for all  $k \geq 1$ , by induction on  $k$  from  $\mathbf{C}\mathbf{A} = \mathbf{A}\mathbf{C}$ . Hence

$$\begin{aligned} [\mathbf{A}, \mathbf{B}\mathbf{C}^{k-1}] &= \mathbf{A}\mathbf{B}\mathbf{C}^{k-1} - \mathbf{B}\mathbf{C}^{k-1}\mathbf{A} = \mathbf{A}\mathbf{B}\mathbf{C}^{k-1} - \mathbf{B}\mathbf{A}\mathbf{C}^{k-1} \\ &= (\mathbf{A}\mathbf{B} - \mathbf{B}\mathbf{A})\mathbf{C}^{k-1} = \mathbf{C}^k. \end{aligned}$$

By Proposition 7.5.1 (c),  $\text{tr } \mathbf{C}^k = 0$ .

- (b) By Theorem 5.7.2 there is a non-zero  $f \in F[x]$  with  $f(\mathbf{C}) = 0$ . Let  $t \geq 0$  be the largest integer with  $x^t$  dividing  $f$ , and write  $f = x^t g$ ; then  $x \nmid g$ , so  $g(0) \neq 0$  by Theorem 5.1.3. Put  $s = t + 1 \geq 1$ . By Theorem 0.9.9,  $\mathbf{C}^s g(\mathbf{C}) = \mathbf{C} f(\mathbf{C}) = 0$ .
- (c) Let  $\mathbf{P}$  have columns  $\mathbf{p}_1, \dots, \mathbf{p}_n$ , where  $(\mathbf{p}_1, \dots, \mathbf{p}_r)$  is a basis of  $U$ ;  $\mathbf{P}$  is invertible by Theorem 2.4.4. For any  $k \geq 1$ , column  $j$  of  $\mathbf{P}^{-1}\mathbf{C}^k\mathbf{P}$  is  $\mathbf{P}^{-1}(\mathbf{C}^k\mathbf{p}_j)$ , the vector  $\mathbf{c}$  with  $\mathbf{P}\mathbf{c} = \mathbf{C}^k\mathbf{p}_j$ , that is, the coefficients of  $\mathbf{C}^k\mathbf{p}_j$  in terms of  $\mathbf{p}_1, \dots, \mathbf{p}_n$  (Theorem 0.9.3). If  $\mathbf{y} \in U$ , then  $\mathbf{y} = \mathbf{C}^s\mathbf{z}$  and  $\mathbf{C}\mathbf{y} = \mathbf{C}^s(\mathbf{C}\mathbf{z}) \in U$ , so  $\mathbf{C}$  maps  $U$  into  $U$ ; hence for  $j \leq r$  the vector  $\mathbf{C}\mathbf{p}_j$  is a combination of  $\mathbf{p}_1, \dots, \mathbf{p}_r$ , and column  $j$  of  $\mathbf{P}^{-1}\mathbf{C}\mathbf{P}$  has zeros below row  $r$ . This gives the block form. By Theorem 7.1.1 and induction on  $k$ ,  $(\mathbf{P}^{-1}\mathbf{C}\mathbf{P})^k = \begin{pmatrix} \mathbf{D}^k & * \\ 0 & \mathbf{N}^k \end{pmatrix}$ , and by Proposition 3.7.6 (c) this is

$$\mathbf{P}^{-1}\mathbf{C}^k\mathbf{P}; \text{ more generally } \mathbf{P}^{-1}q(\mathbf{C})\mathbf{P} = \begin{pmatrix} q(\mathbf{D}) & * \\ 0 & q(\mathbf{N}) \end{pmatrix} \text{ for every } q \in F[x].$$

For  $k = s$ , every vector  $\mathbf{C}^s\mathbf{p}_j$  lies in  $U$ , so every column of  $\mathbf{P}^{-1}\mathbf{C}^s\mathbf{P}$  is zero below row  $r$ , and  $\mathbf{N}^s = 0$ . For  $q = g$ : for  $j \leq r$ ,  $\mathbf{p}_j \in U$  is  $\mathbf{C}^s\mathbf{z}$  for some  $\mathbf{z}$ , so  $g(\mathbf{C})\mathbf{p}_j = g(\mathbf{C})\mathbf{C}^s\mathbf{z} = \mathbf{0}$  by (b) and Theorem 0.9.9. So the first  $r$  columns of  $\mathbf{P}^{-1}g(\mathbf{C})\mathbf{P}$  are zero, and in particular  $g(\mathbf{D}) = 0$ .

- (d) From  $g(\mathbf{D}) = g_0\mathbf{I}_r + \mathbf{D}h(\mathbf{D}) = 0$  we get  $\mathbf{I}_r = \mathbf{D}h_1(\mathbf{D})$  with  $h_1 = -g_0^{-1}h$ . Since

polynomials in  $\mathbf{D}$  commute (Theorem 0.9.9),

$$\mathbf{I}_r = (\mathbf{D} h_1(\mathbf{D}))^s = \mathbf{D}^s h_1(\mathbf{D})^s = \sum_{k=s}^d c_k \mathbf{D}^k$$

for some  $c_k \in F$ , as every term of  $x^s h_1(x)^s$  has degree at least  $s$ . For  $k \geq s$ ,  $\mathbf{N}^k = \mathbf{N}^s \mathbf{N}^{k-s} = 0$ , so by (c), Theorem 3.7.5 and Definition 0.9.12,  $\text{tr } \mathbf{C}^k = \text{tr}(\mathbf{P}^{-1} \mathbf{C}^k \mathbf{P}) = \text{tr } \mathbf{D}^k + \text{tr } \mathbf{N}^k = \text{tr } \mathbf{D}^k$ . Taking traces,

$$r \cdot 1 = \text{tr } \mathbf{I}_r = \sum_{k=s}^d c_k \text{tr } \mathbf{D}^k = \sum_{k=s}^d c_k \text{tr } \mathbf{C}^k = 0$$

by (a). But  $r \geq 1$  and  $F$  has characteristic 0, so  $r \cdot 1 \neq 0$ , a contradiction.

- (e) By (d),  $r = 0$ , so  $\text{col}(\mathbf{C}^s) = \{\mathbf{0}\}$  and  $\mathbf{C}^s = 0$ . For example, with  $\mathbf{A} = \mathbf{E}_{12} + \mathbf{E}_{23}$  and  $\mathbf{B} = \text{diag}(0, 1, 2)$  in  $M_3(\mathbb{Q})$ ,  $\mathbf{C} = [\mathbf{A}, \mathbf{B}] = \mathbf{E}_{12} + \mathbf{E}_{23} = \mathbf{A}$ , which commutes with  $\mathbf{A}$ , and indeed  $\mathbf{C}^3 = 0$ . In Chapter 8 the statement “ $\text{tr } \mathbf{C}^k = 0$  for all  $k \geq 1$ ” forces nilpotency in characteristic 0” becomes a statement about eigenvalues.